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On Discovery of Local Independence over Continuous Variables via Neural Contextual Decomposition

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Abstract

Conditional independence provides a way to understand causal relationships among the variables of interest. An underlying system may exhibit more fine-grained causal relationships especially between a variable and its parents, which will be called the local independence relationships. One of the most widely studied local relationships is Context-Specific Independence (CSI), which holds in a specific assignment of conditioned variables. However, its applicability is often limited since it does not allow continuous variables: data conditioned to the specific value of a continuous variable contains few instances, if not none, making it infeasible to test independence. In this work, we define and characterize the local independence relationship that holds in a specific set of joint assignments of parental variables, which we call context-set specific independence (CSSI). We then provide a canonical representation of CSSI and prove its fundamental properties. Based on our theoretical findings, we cast the problem of discovering multiple CSSI relationships in a system as finding a partition of the joint outcome space. Finally, we propose a novel method, coined neural contextual decomposition (NCD), which learns such partition by imposing each set to induce CSSI via modeling a conditional distribution. We empirically demonstrate that the proposed method successfully discovers the ground truth local independence relationships in both synthetic dataset and complex system reflecting the real-world physical dynamics.

Keywords: Context-Specific Independence, Local Independence, Causal Discovery

1. Introduction

Discovering the causal relationships in a system is an important and challenging problem in many areas of scientific research such as social science (Sobel, 1995), biology (Shipley, 2016), and economics (Angrist et al., 1996; Angrist and Pischke, 2008; Imbens and Rubin, 2015; Banerjee et al., 2016; Imbens, 2020). There have been many causal discovery algorithms finding causal relationships given observational data (Spirtes et al., 2000; Chickering, 2002; Hoyer et al., 2008; Shimizu et al., 2006; Zhang et al., 2011; Zheng et al., 2018; Zhu et al., 2019). These methods

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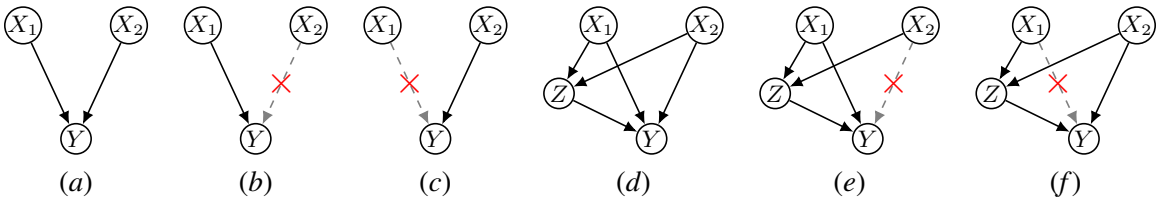


Figure 1: (a) Causal graph in Example 1. (b, c) CSI can represent the local independence of Y and X_2, but not the other. In contrast, CSSI is able to represent both. (d) Augmented causal graph with PIV Z added in Example 4. (e, f) PIV representing CSSIs for each context set.

often exploit conditional independence either explicitly (constraint-based methods) or implicitly (score-based or gradient-based).

A system often exhibits more fine-grained relationships between a variable and its parents, i.e., a local independence relationship. For example, when pushing an object on the ground, it will move *only* when a force exceeds the friction which is determined by the mass and the ground. Context-specific independence (CSI) (Poole and Zhang, 2003; Boutilier et al., 2013) is the independence that holds in a *certain* conditioning value (i.e., *context*) as opposed to *any* value of the conditioned variables. It has been shown that such independence leads to more efficient probabilistic inference by exploiting the underlying local structure (Poole, 1998; Poole and Zhang, 2003; Gogate and Domingos, 2010; Van den Broeck et al., 2011; Dal et al., 2018). Further, it allows the identification of causal effects, which would not be possible without CSI relationships (Tikka et al., 2019; Robins et al., 2020).

Despite the fact that many real-world scenarios involve continuous variables, most prior works on local independence relationships assumed that variables are discrete. This is partly due to the notion of CSI that is inherently suited for discrete variables as conditioning on the specific value of a continuous variable is infeasible, where the resulting subset of data would be practically empty. Hence, it is non-trivial to explore how local independence for continuous variables can be empirically discovered and utilized in probabilistic or causal inferential tasks.

Consider a system as an example consisting of three observed variables X_1, X_2, Y, and an unobserved one U. We describe the system using a structural causal model (SCM) (Pearl, 2009; Peters et al., 2017). Let X_1, X_2 follow a uniform distribution between 0 and 1, and U follow a standard Gaussian distribution. Further, let Y be determined as $Y = X_1 + U$ if $X_1 X_2 < 1/2$ and $Y = X_2 + U$ otherwise. Focusing on the observable variables, while Y depends on both X_1 and X_2 overall (Fig. 1(*a*)), Y depends only (i) on X_1 under a certain condition and (ii) on X_2, otherwise. Since $X_1 < 1/2$ implies the condition $X_1 X_2 < 1/2$ and Y being the function of X_1 and U, it can be represented as CSI (Fig. 1(*b*)). On the other hand, CSI cannot capture the other local independence where Y is determined only by X_2 and U since Y and X_1 are always dependent given any assignment of X_2.

Against this background, we define context-set specific independence (CSSI), a generalized notion of local independence where conditional independence holds in a certain *context set*, a set of joint assignments of the variables. We characterize context-set specific independence in order to uncover CSSIs for continuous variables, which is more challenging due to an infinite number of context sets. Our characterization leads to the design of NCD (Neural Contextual Decomposition) to

find the partitions of the joint outcome space where each partition set induces a CSSI. Our approach is based on approximating a conditional distribution for each partition given a subset of parents. For instance, based on the above example, our method learns two partitions of the Cartesian product of the domains of X_1 and X_2, where each partition is flexibly modeled as $\hat{P}(Y \mid X_1, Z)$ and $\hat{P}(Y \mid X_2, Z)$.

Our contributions are summarized as follows. (i) We introduce the notion of context-set specific independence (CSSI), a new class of local conditional independence relationships. This generalizes two well-known local independence, CSI and partial conditional independence (PCI), and permits continuous conditioned variables. (ii) We characterize context-set specific independence by relating possible CSSIs arisen from a given distribution. We examine a canonical representation of CSSIs and provide a sufficient condition under which a unique CSSIs exists. These characterizations lead to defining contextual decomposition, a way to understand an entire group of CSSIs presenting in a given dataset. (iii) We devise neural contextual decomposition, a simple and effective method for finding a contextual decomposition without directly testing for individual local independence. It utilizes an auxiliary variable (partition indicator variable) used for training of a neural network for context decomposition. Empirical evaluations on a synthetic dataset and Spriteworld demonstrate the effectiveness of the proposed method in recovering local independence relationships.

2. Preliminaries

Throughout this paper, we use capital letters for random variables and small letters for their assignments. Bold letters denote the set of random variables or its assignments. Calligraphy letters can be the domain of corresponding variables or complex mathematical objects.

2.1. Structural Causal Model

We adopt a structural causal model (SCM) (Pearl, 2009) as a causal framework to understand generated data. An SCM $\mathcal{M}$ is defined as a tuple $\langle \mathbf{V}, \mathbf{U}, \mathbf{F}, P(\mathbf{U}) \rangle$, where $\mathbf{V} = \{V_1, \cdots, V_d\}$ and $\mathbf{U}$ are a set of endogenous and exogenous variables, respectively. $\mathbf{F} = \{f_1, \cdots, f_d\}$ is a set of functions determining each endogenous variable, i.e., $V_j \leftarrow f_j(Pa(j), \mathbf{U}_j)$ where $Pa(j) \subseteq \mathbf{V} \setminus \{V_j\}$ is a parent of V_j and $\mathbf{U}_j \subseteq \mathbf{U}$. We may use variable V_i and its index i interchangeably. $\mathcal{V}_j$ is the outcome space of each variable V_j. In this work, we restrict to a Markovian model[1] and assume that an SCM satisfies structural minimality (Pearl, 2009; Peters et al., 2017), which asserts that $j \in Pa(i)$ if and only if there exist two values for j that lead to different values for i, i.e., $f_i(v_j, \mathbf{v}_{Pa(i)\setminus\{j\}}, \mathbf{u}_i) \neq f_i(v'_j, \mathbf{v}_{Pa(i)\setminus\{j\}}, \mathbf{u}_i)$ where $v_j, v'_j \in \mathcal{V}_j$ and $\mathbf{v}_{Pa(i)\setminus\{j\}} \in \mathcal{V}_{Pa(i)\setminus\{j\}}$. An SCM $\mathcal{M}$ induces a causal graph $\mathcal{G} = (\mathbf{V}, \mathbf{E})$, which is a directed acyclic graph (DAG), where $\mathbf{V} = \{1, \ldots, d\}$ and $\mathbf{E} \subseteq \mathbf{V} \times \mathbf{V}$ are the set of nodes and edges, respectively. If $i \in Pa(j)$ in $\mathcal{M}$, then a directed edge (i, j) is in $\mathbf{E}$. In the language of SCM, each node i corresponds to a random variable V_i and each edge (i, j) denotes a direct causal relationship from V_i to V_j.

2.2. Local Independence Relationship of Discrete Variables

As described earlier, in many cases, the causal mechanism exhibits fine-grained independence relationships, which do not hold in general but only under certain conditions, i.e., *local independence*.

1. Conditional independence is a statistical notion and would be irrelevant to whether the model is Markovian or semi-Markovian. However, Markovian models will allow clearer interpretation of context-specific independence with respect to the functional form of the target variable irrespective of regimes, i.e., observational or interventional.

To begin with, we first provide the definition of context-specific independence (CSI) and partial conditional independence (PCI). We adopt the definitions of CSI and PCI from prior works (Boutilier et al., 2013; Pensar et al., 2015, 2016) for the relationship between a variable and its parents. For the definitions of CSI and PCI, variables are assumed to be discrete. Let $\mathbf{X}$ be a non-empty set of parents of Y and $\mathbf{X}_A, \mathbf{X}_B$ be non-empty partitions of $\mathbf{X}$, i.e., $\mathbf{X}$ is disjoint union of $\mathbf{X}_A$ and $\mathbf{X}_B$.

Definition 1 (Context-Specific Independence (CSI)) *Y is said to be* contextually independent *of $\mathbf{X}_B$ given the context $\mathbf{X}_A = \mathbf{x}_A$ if $P(y \mid \mathbf{x}_A, \mathbf{x}_B) = P(y \mid \mathbf{x}_A)$, holds for all $y \in \mathcal{Y}$ and $\mathbf{x}_B \in \mathcal{X}_B$ whenever $P(\mathbf{x}_A, \mathbf{x}_B) > 0$. This will be denoted by $Y \perp\!\!\!\perp \mathbf{X}_B \mid \mathbf{X}_A = \mathbf{x}_A$.*

Definition 2 (Partial Conditional Independence (PCI)) *Y is said to be* partially conditionally independent *of $\mathbf{X}_B$ in the domain $\mathcal{D}_B \subseteq \mathcal{X}_B$ given the context $\mathbf{X}_A = \mathbf{x}_A$ if $P(y \mid \mathbf{x}_A, \mathbf{x}_B) = P(y \mid \mathbf{x}_A, \mathbf{x}'_B)$, holds for all $y \in \mathcal{Y}$ and $\mathbf{x}_B, \mathbf{x}'_B \in \mathcal{D}_B$ whenever $P(\mathbf{x}_A, \mathbf{x}_B) > 0$ and $P(\mathbf{x}_A, \mathbf{x}'_B) > 0$. This will be denoted by $Y \perp\!\!\!\perp \mathbf{X}_B \mid \mathcal{D}_B, \mathbf{X}_A = \mathbf{x}_A$.*

PCI is a more fine-grained notion than CSI and, thus, PCI does not imply CSI: $P(y \mid \mathbf{x}_A, \mathbf{x}_B) = P(y \mid \mathbf{x}_A, \mathbf{x}'_B)$ for all $\mathbf{x}_B, \mathbf{x}'_B \in \mathcal{D}_B$ is not equivalent to $P(y \mid \mathbf{x}_A, \mathbf{x}_B) = P(y \mid \mathbf{x}_A)$. Further discussions of the related works on local independence relationships are provided in Sec. A.1 and 6.

3. Compactly Representing Local Independence of Continuous Variables

Our goal is to find such local independence relationships in a system that may contain continuous variables. The information about local independence relationships of continuous variables can also be leveraged in various tasks as demonstrated in the case of discrete variables. Given that a naive generalization of CSI or PCI to continuous variables seems practically implausible, we first define an alternative notion of local independence, which we call context-set specific independence (CSSI). Then, we characterize CSSI and describe how it can be equipped within the framework of SCM. All omitted proofs can be found in the Appendix.

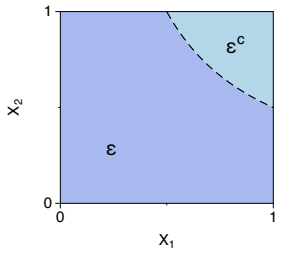


Figure 2: Example 1.

3.1. Local Independence Relationship over Continuous Variables

Here, we focus on a causal mechanism of a target variable Y and its parents $\mathbf{X} = \{X_1, \cdots, X_d\}$, i.e., SCM $\mathcal{M} = \langle \mathbf{V}, \mathbf{U}, \mathbf{F}, P(\mathbf{U}) \rangle$ where $\{X_1, \cdots, X_d, Y\} \subseteq \mathbf{V}$ is a set of continuous random variables and $Y = f(\mathbf{X}, U)$. We denote $\mathbf{X}_A = \{X_i \mid i \in A\}$ where $A \subseteq Pa(Y) = \{1, \cdots, d\}$. Similarly, $\mathbf{X}_{A^c} = \{X_i \mid i \in A^c\}$ where $A^c = Pa(Y) \setminus A$. Throughout the continuous-variable analysis, densities are taken with respect to Lebesgue measure and are strictly positive on the outcome space. Equalities of conditional distributions on a context are understood almost surely under the distribution restricted to that context; changing a conditional kernel on a null set does not change the independence relationship. We first define a context-set specific independence (CSSI).

Definition 3 (Context-Set Specific Independence) *Let $\mathbf{X} = \{X_1, \cdots, X_d\}$ be a non-empty set of the parents of Y in a causal graph, and $\mathcal{E} \subseteq \mathcal{X}$ be an event of positive probability. $\mathcal{E}$ is a* **context set** *inducing* **context-set specific independence (CSSI)** *of $\mathbf{X}_{A^c}$ from Y if there exists a conditional density q such that $p(y \mid \mathbf{x}) = q(y \mid \mathbf{x}_A)$ for $P_{\mathbf{X}|\mathcal{E}}$-almost every $\mathbf{x}$, as an equality of conditional distributions of Y. We denote this by $Y \perp\!\!\!\perp \mathbf{X}_{A^c} \mid \mathbf{X}_A, \mathcal{E}$.*

CSSI extends CSI and PCI in the sense that the independence holds in a set of conditioned values in CSSI since they consider a point-based condition. We revisit the earlier example to illustrate how CSI, PCI, and CSSI (Defs 1 to 3) represent the local independence relationships in the system (insufficiency results for CSI and PCI in representing Example 1 is depicted in Sec. A.1.)

Example 1 *Let X_1, X_2 be independent Unif $[0, 1]$ variables and let $U \sim \mathcal{N}(0, 1)$ be independent of them. Let Y be:*

$$Y = \begin{cases} X_1 + U & \text{if} \quad X_1 X_2 < 1/2, \\ X_2 + U & \text{if} \quad X_1 X_2 \geq 1/2. \end{cases}$$

Let $\mathcal{E} = \{(x_1, x_2) \mid x_1 x_2 < 1/2\}$. Then, $Y \perp\!\!\!\perp X_2 \mid X_1, \mathcal{E}$ and $Y \perp\!\!\!\perp X_1 \mid X_2, \mathcal{E}^c$ hold. On the other hand, $Y \not\perp\!\!\!\perp X_1 \mid X_2 = c$ for any c.

As described earlier, we focus on the discovery of local independence (i.e., CSSI) of continuous variables and henceforth assume variables are continuous. In a case where the variables are discrete, CSSI can be discovered by first (i) discovering PCI relationships and then (ii) integrating them (see Sec. A.1 for the detail). We now introduce important properties of CSSI.

Proposition 4 (CSSI Entailment) *Suppose a CSSI relationship $Y \perp\!\!\!\perp \mathbf{X}_{A^c} \mid \mathbf{X}_A, \mathcal{E}$ holds. Then, the following CSSI relationships also hold:*

(i) $Y \perp\!\!\!\perp \mathbf{X}_{B^c} \mid \mathbf{X}_B, \mathcal{E}$ for any $B \supseteq A$, (ii) $Y \perp\!\!\!\perp \mathbf{X}_{A^c} \mid \mathbf{X}_A, \mathcal{F}$ for any $\mathcal{F} \subseteq \mathcal{E}$ with $P(\mathcal{F}) > 0$.

Due to such implications, we are interested in CSSI with *minimal* conditioned variables.

Definition 5 *A CSSI of the form $Y \perp\!\!\!\perp \mathbf{X}_{A^c} \mid \mathbf{X}_A, \mathcal{E}$ is said to be **regular** if there does not exist any $B \subsetneq A$ such that $Y \perp\!\!\!\perp \mathbf{X}_{B^c} \mid \mathbf{X}_B, \mathcal{E}$ holds. Then, we say $\mathbf{X}_A$ to be a **local parent set** of Y in $\mathcal{E}$.*[2]

Among the CSSIs with the same context set, the regular CSSIs are the most *informative*, in the sense that the set of the conditioned variables cannot be further reduced. We say a CSSI is **trivial** if $A = Pa(Y)$ (i.e., $A^c = \emptyset$ and $\mathbf{X} = \mathbf{X}_A$), since it trivially holds for any $\mathcal{E} \subseteq \mathcal{X}$. A trivial CSSI on $\mathcal{E} = \mathcal{X}$ is regular when no proper subset of $Pa(Y)$ suffices to represent the conditional distribution, i.e., under statistical causal minimality. Structural minimality of the SCM alone does not imply this condition. In general, regular CSSI may not be unique on the given context set. We show the non-uniqueness of local parent sets by an example below.

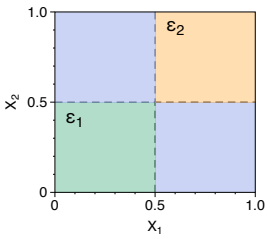


Figure 3: Example 2.

Example 2 *Let X_1, X_2, X_3 be independent Unif $[0, 1]$ variables and let $U \sim \mathcal{N}(0, 1)$ be independent of them. Let Y be:*

$$Y = \begin{cases} X_3 + U & \text{if} \quad X_1, X_2 < 1/2, \\ X_3 + 2 \cdot U & \text{if} \quad X_1, X_2 \geq 1/2, \\ X_1 + X_2 + X_3 + U & \text{otherwise}. \end{cases}$$

2. We use $\mathbf{pa}^{\mathcal{E}}$ and $Pa^{\mathcal{E}}(Y)$ to denote the local parent set of Y in $\mathcal{E}$, i.e., $\mathbf{pa}^{\mathcal{E}} = \mathbf{X}_A$ and $Pa^{\mathcal{E}}(Y) = A$.

Let $\mathcal{E}_1 = \{\mathbf{x} \mid x_1, x_2 < 1/2\}$, $\mathcal{E}_2 = \{\mathbf{x} \mid x_1, x_2 \geq 1/2\}$. *Then, regular CSSIs* $Y \perp\!\!\!\perp X_{12} \mid X_3, \mathcal{E}_1$ *and* $Y \perp\!\!\!\perp X_{12} \mid X_3, \mathcal{E}_2$ *hold. Also, the following regular CSSIs hold on* $\mathcal{E}_1 \cup \mathcal{E}_2$*:*

$$Y \perp\!\!\!\perp X_1 \mid X_{23}, \mathcal{E}_1 \cup \mathcal{E}_2 \qquad \textit{and} \qquad Y \perp\!\!\!\perp X_2 \mid X_{13}, \mathcal{E}_1 \cup \mathcal{E}_2.$$

In the example above, there exist two distinct regular CSSIs on $\mathcal{E}_1 \cup \mathcal{E}_2$. If the local parent set is not unique on the given context set, discovering the local parent set would yield an inconsistent result. Thus, we explore a sufficient condition that guarantees the uniqueness of the local parent set. We now provide an intersection property of CSSI, which will be used to derive a sufficient condition.

Proposition 6 (Intersection Property of CSSI) *Suppose* $\mathcal{E} \subseteq \mathcal{X}$ *is convex and has positive probability. If CSSI relationships* $Y \perp\!\!\!\perp \mathbf{X}_{A^c} \mid \mathbf{X}_A, \mathcal{E}$ *and* $Y \perp\!\!\!\perp \mathbf{X}_{B^c} \mid \mathbf{X}_B, \mathcal{E}$ *hold, then* $Y \perp\!\!\!\perp \mathbf{X}_{(A\cap B)^c} \mid \mathbf{X}_{A\cap B}, \mathcal{E}$ *hold.*

In Example 2, we cannot derive $Y \perp\!\!\!\perp X_{12} \mid X_3, \mathcal{E}_1 \cup \mathcal{E}_2$ from the two regular CSSIs on $\mathcal{E}_1 \cup \mathcal{E}_2$ as $\mathcal{E}_1 \cup \mathcal{E}_2$ is not convex. We now introduce a sufficient condition that guarantees the uniqueness of regular CSSIs given a context set $\mathcal{E}$.

Theorem 7 (Uniqueness of Local Parent Set) *Suppose* $\mathcal{E} \subseteq \mathcal{X}$ *is convex and has positive probability. There exists a unique* $A \subseteq Pa(Y)$ *such that the regular CSSI relationship* $Y \perp\!\!\!\perp \mathbf{X}_{A^c} \mid \mathbf{X}_A, \mathcal{E}$ *holds.*

Proof An inclusion-minimal sufficient subset exists because $Pa(Y)$ is finite. Suppose regular CSSIs $Y \perp\!\!\!\perp \mathbf{X}_{A^c} \mid \mathbf{X}_A, \mathcal{E}$ and $Y \perp\!\!\!\perp \mathbf{X}_{B^c} \mid \mathbf{X}_B, \mathcal{E}$ hold for some A and B. By Prop. 6 and the convexity of $\mathcal{E}$, $(Y \perp\!\!\!\perp \mathbf{X}_{(A\cap B)^c} \mid \mathbf{X}_{A\cap B}, \mathcal{E})$ holds true. Due to the minimality (i.e., regular), $A \cap B$ cannot be a proper subset of A nor B. Hence, $A = B = A \cap B$. ■

This implies that one may devise a conditional independence test or causal discovery algorithm to find the (unique) local parent set on the convex subset of data (e.g., a rectangular $\mathcal{E} = [a_1, b_1] \times \cdots \times [a_d, b_d] \in \mathbb{R}^d$ or a ball $\mathcal{E} = \{\mathbf{x} \mid \|\mathbf{x} - \mathbf{x}_0\|_2 \leq r\}$). If the subset of data is non-convex, the local parent set *may* not be unique,[3] and thus the results may not be valid without any additional assumptions. We provide a more general condition for Prop. 6 and Thm. 7 in Sec. B.

4. Discovering Local Independence Relationships by Learning the Partition

We investigate the representation and discovery of *multiple* CSSIs in a system. To compactly represent CSSIs, we introduce a decomposition of the parents' outcome space, where each partition corresponds to a CSSI-inducing context set. Then, we transform the finding of such decomposition as a learning problem involving an auxiliary variable representing the decomposition. Finally, we develop a neural approach to learning the decomposition via the auxiliary variable.

4.1. Representing CSSIs in a System as a Partition

We have seen through examples that a system may exhibit multiple CSSIs. We define a canonical notion of CSSI in order to examine a compact representation of CSSIs.

3. Note that this does not imply that local parent set is *always* not unique on the non-convex subset of data, e.g., $Y \perp\!\!\!\perp X_2 \mid X_1, \mathcal{E}$ is the unique regular CSSI on the non-convex set $\mathcal{E}$ in Example 1.

Definition 8 *A CSSI relationship $Y \perp\!\!\!\perp \mathbf{X}_{A^c} \mid \mathbf{X}_A, \mathcal{E}$ is **canonical** if there does not exist any $\mathcal{F} \subseteq \mathcal{E}$ of positive probability such that $Y \perp\!\!\!\perp \mathbf{X}_{B^c} \mid \mathbf{X}_B, \mathcal{F}$ and $B \subsetneq A$.*

By definition, the canonicality of a CSSI implies its regularity. Canonical CSSIs are the most fine-grained ones since any subset of the context set also entails the same regular CSSI relationship. In Example 2, CSSI relationship $Y \perp\!\!\!\perp X_1 \mid X_{23}, \mathcal{E}_1 \cup \mathcal{E}_2$ is regular but not canonical since $Y \perp\!\!\!\perp X_{12} \mid X_3, \mathcal{E}_1$. If a trivial CSSI on $\mathcal{E} = \mathcal{X}$ is canonical, a system does not exhibit any CSSI relationship (e.g., Example 3 in Sec. D). We now define a compact representation of multiple CSSIs in a system.

Definition 9 (Contextual Decomposition) *Let $\{\mathcal{E}_k\}_{k=0}^N$ be a partition of $\mathcal{X}$. $\{(\mathcal{E}_k, A_k)\}_{k=0}^N$ is a **contextual decomposition (CD)** if regular CSSI $Y \perp\!\!\!\perp \mathbf{X}_{A_k^c} \mid \mathbf{X}_{A_k}, \mathcal{E}_k$ holds on every positive-probability cell, where $A_0 = Pa(Y)$, $A_k \subsetneq Pa(Y)$ and $P(\mathcal{E}_k) > 0$ for all $k \geq 1$. The residual $\mathcal{E}_0$ may have probability zero, in which case it imposes no regularity or independence condition. We say a CD is **canonical** if its CSSIs on all positive-probability cells are canonical, and **distinctive** if $A_i \neq A_j$ for all $i \neq j$.*

$\mathcal{E}_0$ is a set to cover a remaining subset of $\mathcal{X}$ excluding non-trivial context sets. $(\mathcal{E}_0, A_0)$ corresponds to a trivial CSSI. When the full parent set is statistically minimal, $\{(\mathcal{X}, Pa(Y))\}$ is a CD, called the **trivial CD**. When a system does not exhibit any CSSI relationship, a trivial CD is the only existing CD. In Example 2, both $\{(\mathcal{E}_0, \mathbf{X}_{123}), (\mathcal{E}_1, \mathbf{X}_3), (\mathcal{E}_2, \mathbf{X}_3)\}$ and $\{(\mathcal{E}_0, \mathbf{X}_{123}), (\mathcal{E}_1 \cup \mathcal{E}_2, \mathbf{X}_{23})\}$ are CDs where $\mathcal{E}_0 = (\mathcal{E}_1 \cup \mathcal{E}_2)^c$. The former is canonical, but the latter is not. Contextual decomposition can be viewed as the discretization of the joint outcome space, in contrast to discretizing each variable which is a commonly used strategy to handle continuous variables (Nyman et al., 2017). As a canonical CSSI is a fine-grained notion of CSSI relationship, a canonical CD can also be viewed as a fine-grained CD. While a canonical CD may not be unique (e.g., Example 5 in Sec. D), the following theorem characterizes the shared property of the canonical CDs in a system.

Theorem 10 *Let $\{(\mathcal{E}_i, A_i)\}_{i=0}^N$ and $\{(\mathcal{F}_j, B_j)\}_{j=0}^M$ be canonical CDs where $\mathcal{E}_i$ and $\mathcal{F}_j$ are open sets for all $i, j \geq 1$. Then, the following holds: (i) if $p(\mathcal{E}_i \cap \mathcal{F}_j) > 0$ then $A_i = B_j$, and (ii) for any $C \subseteq Pa(Y)$, $p(\mathcal{E}[C] \triangle \mathcal{F}[C]) = 0$[4] where $\mathcal{E}[C] = \bigcup_{A_i = C} \mathcal{E}_i$ and $\mathcal{F}[C] = \bigcup_{B_j = C} \mathcal{F}_j$.*

Roughly, any intersecting context sets (i.e., $p(\mathcal{E}_i \cap \mathcal{F}_j) > 0$) from different canonical CDs share the same local parent set (i.e., $A_i = B_j$). Also, for any $C \subseteq Pa(Y)$, the union of the context sets having the same local parent set C is identical for any canonical CSSI in a system (i.e., $p(\mathcal{E}[C] \triangle \mathcal{F}[C]) = 0$). We provide an example of canonical CDs to elaborate Thm. 10 in Example 5 (Sec. D). An interesting property of CDs is that the union of the context sets having the same local parents set may no longer entail the same one, i.e., $\mathbf{pa}^{\mathcal{E}_1 \cup \mathcal{E}_2} \neq \mathbf{pa}^{\mathcal{E}_1} \cup \mathbf{pa}^{\mathcal{E}_2}$, in general as shown in Example 2. For the case of distinctive canonical CDs, we have stronger uniqueness which directly follows from Thm. 10.

Corollary 11 (Uniqueness of Distinctive Canonical CD) *Let $\{(\mathcal{E}_i, A_i)\}_{i=0}^N$ and $\{(\mathcal{F}_j, B_j)\}_{j=0}^M$ be distinctive canonical CDs where $\mathcal{E}_i$ and $\mathcal{F}_j$ are open sets for all $i, j \geq 1$. Then, the following hold: (i) $N = M$, and (ii) if $p(\mathcal{E}_i \cap \mathcal{F}_j) > 0$ then $A_i = B_j$ and $p(\mathcal{E}_i \triangle \mathcal{F}_j) = 0$.*

In words, distinctive canonical CD is *unique* since any intersecting context sets (i.e., $p(\mathcal{E}_i \cap \mathcal{F}_j) > 0$) from different distinctive canonical CDs share the same local parent set (i.e., $A_i = B_j$), and their difference is negligible (i.e., $p(\mathcal{E}_i \triangle \mathcal{F}_j) = 0$).

4. $\triangle$ is a symmetric difference, i.e., $A \triangle B = (A \setminus B) \cup (B \setminus A)$

4.2. Representing Contextual Decomposition with an Auxiliary Variable

We now introduce a partition indicator variable, which will serve as a mapping between $\mathcal{X}$ and an arbitrary contextual decomposition. This variable will be useful in transforming the task of testing for CSSI to the task of learning such decomposition. We first show that a CD can be represented by introducing an auxiliary variable.

Proposition 12 (Expressing CSSI as CSI with Auxiliary Variable) *Let $\{\mathcal{E}_k\}_{k=0}^N$ partition $\mathcal{X}$, let $A_0 = Pa(Y)$ and $A_k \subsetneq Pa(Y)$ for $k \geq 1$, and define $Z = z_k$ when $\mathbf{X} \in \mathcal{E}_k$. On each positive-probability cell, $Y \perp\!\!\!\perp \mathbf{X}_{A_k^c} \mid \mathbf{X}_{A_k}, \mathcal{E}_k$ holds if and only if $Y \perp\!\!\!\perp \mathbf{X}_{A_k^c} \mid \mathbf{X}_{A_k}, Z = z_k$ holds. Consequently, the given pairs form a contextual decomposition if and only if these CSI relationships hold and each A_k is inclusion-minimal among their valid conditioning sets on every positive-probability cell, with $P(\mathcal{E}_k) > 0$ for $k \geq 1$.*

In Example 1, we can introduce Z so that $Y \perp\!\!\!\perp X_2 \mid X_1, Z = z_1$ and $Y \perp\!\!\!\perp X_1 \mid X_2, Z = z_2$ where $Z = z_1$ if $\mathbf{X} \in \mathcal{E}$ and z_2 otherwise. We now formally define a partition indicator variable.

Definition 13 *Let $\{\mathcal{E}_k\}_{k=0}^N$ be a partition of $\mathcal{X}$ and a random variable Z be defined as $Z = z_k$ if $\mathbf{X} \in \mathcal{E}_k$ for all $0 \leq k \leq N$. Variable Z is a **partition indicator variable (PIV)** if for every $k \geq 1$ with $P(\mathcal{E}_k) > 0$ there exists some $A_k \subsetneq Pa(Y)$ such that $Y \perp\!\!\!\perp \mathbf{X}_{A_k^c} \mid \mathbf{X}_{A_k}, Z = z_k$.*

PIV can be viewed as a particular type of sufficient set of statistics.[5] Prop. 12 expresses the regionwise CSSIs of a PIV. To obtain a CD, choose inclusion-minimal parent sets on the positive-probability cells and require the residual full-parent set to be regular whenever $P(\mathcal{E}_0) > 0$; the CSI relationships alone do not enforce this minimality. With PIV Z, each CSSI relationship $Y \perp\!\!\!\perp \mathbf{X}_{A_k^c} \mid \mathbf{X}_{A_k}, \mathcal{E}_k$ is equivalently expressed as $Y \perp\!\!\!\perp \mathbf{X}_{A_k^c} \mid \mathbf{X}_{A_k}, Z = z_k$. Thus, for $P_{\mathbf{X}|\mathcal{E}_k}$-almost every $\mathbf{x}$,

$$p(y \mid \mathbf{x}) = p(y \mid \mathbf{x}, z_k) = p(y \mid \mathbf{x}_{A_k^c}, \mathbf{x}_{A_k}, z_k) = p(y \mid \mathbf{x}_{A_k}, z_k), \tag{1}$$

where the first equality holds by definition, and the last equality holds since PIV entails CSI relationships. Therefore some of the parent variables (i.e., $\mathbf{X}_{A_k^c}$) would be ignored for modeling the conditional distribution of Y given $\mathcal{E}_k$. In contrast to CSI, which uses the value of a proper subset of $Pa(Y)$ ignoring the rest, the value of Z is determined by the whole $\mathbf{X}$, in general, and a subset of $Pa(Y)$ to be ignored (i.e., $\mathbf{X}_{A_k^c}$) could involve in determining the value of Z.

4.3. Neural Contextual Decomposition

As described earlier, conditional independence tests can be used to discover the CSSI relationship on a particular subset of data. Further, assuming strictly positive densities, testing on a convex subset of data is sufficient to obtain a unique local parent set without any additional assumptions. However, it is generally infeasible to test on every possible subset to discover multiple CSSIs in a given system.

Against this background, we propose **Neural Contextual Decomposition (NCD)**, a neural approach to recovering *distinct* contextual decomposition from given data $P(\mathbf{V})$ via learning PIV Z.

5. Chicharro et al. (2020) utilized a sufficient set of statistics with a focus on discovering a causal structure. However, we provided the characterization and representation of local independence and its fundamental properties. Although the concept of a sufficient set of statistics and related rules is valid for continuous variables, their information bottleneck approach is restricted to the discrete variables.

Recalling $Y \perp\!\!\!\perp \mathbf{X}_{A_k^c} \mid \mathbf{X}_{A_k}, Z = z_k$, we let $\mathcal{Z} \subseteq \{0,1\}^d$ so that the local parent set A_k corresponds to the set of the indices of the nonzero element of z_k, e.g., $Y \perp\!\!\!\perp X_2 \mid X_{13}, Z = (1,0,1)$.[6] With PIV Z, we can write the conditional density almost surely as:

$$p(y \mid \mathbf{x}) = \textstyle\sum_z p(y \mid \mathbf{x}, z)p(z \mid \mathbf{x}) = \sum_z p(y \mid \mathbf{x}_{A_z}, z)p(z \mid \mathbf{x}), \tag{2}$$

where A_z is A_k corresponding to z_k and $p(z \mid \mathbf{x}) = \delta(z = z_i)$ if $\mathbf{x} \in \mathcal{E}_i$. Our method models $p(y \mid \mathbf{x}, z)$ and $p(z \mid \mathbf{x})$.

Modeling $p(y \mid \mathbf{x}, z)$ and $p(z \mid \mathbf{x})$. Since $p(y \mid \mathbf{x}, z) = p(y \mid \mathbf{x}_{A_z}, z)$, some of the parent variables (i.e., $\mathbf{x}_{A_z^c}$) are redundant for modeling a conditional distribution. Therefore, our method approximates the conditional distribution $p(y \mid \mathbf{x}, z)$ as $\hat{p}(y \mid \mathbf{x}, z)$ and let a neural network f_θ takes $(\mathbf{x} \odot z, z)$ as an input where $\odot$ denotes an element-wise product and outputs the parameters of the estimator $\hat{p}(y \mid \mathbf{x}, z)$.[7] For the input of f_θ, we simply concatenate $\mathbf{x} \odot z$ and z, e.g., if $z = (1,0,1)$ then f_θ takes $(\mathbf{x} \odot z, z) = (x_1, 0, x_3, 1, 0, 1)$ as an input. We emphasize that labeling (i.e., concatenation of z) is essential since $\mathbf{x}_{A_k^c}$ cannot be ignored without conditioning on z, i.e., $p(y \mid \mathbf{x}) = p(y \mid \mathbf{x}, z_k) = p(y \mid \mathbf{x}_{A_k}, z_k)$ holds almost surely on $\mathcal{E}_k$ but $p(y \mid \mathbf{x}) \neq p(y \mid \mathbf{x}_{A_k})$ and $p(y \mid \mathbf{x}, z_k) \neq p(y \mid \mathbf{x}_{A_k})$ in general. Moreover, without labeling, it is unable to distinguish whether the zero entries of input are either masked or its assignment being zero. We also model the conditional distribution $p(z \mid \mathbf{x})$ as $\hat{p}(z \mid \mathbf{x})$, and let a neural network g_ϕ takes $\mathbf{x}$ as input and outputs the parameters of Bernoulli variables. That is, $g_\phi(\mathbf{x}) = (\pi_1, \cdots, \pi_d)$ and $z_{(k)}|\mathbf{x} \sim \text{Bernoulli}(\pi_k)$ for all k where $z = (z_{(1)}, \cdots, z_{(d)}) \in \{0,1\}^d$. We provide further discussions of the related works on neural network based causal discovery methods in Sec. 6.

Training objective. Our model becomes $p(y \mid \mathbf{x}; \theta, \phi) = \sum_z \hat{p}(y \mid \mathbf{x}, z)\hat{p}(z \mid \mathbf{x}) = \mathbb{E}_{\hat{p}(z|\mathbf{x})}\hat{p}(y \mid \mathbf{x}, z)$, and its maximum likelihood estimation is as follows:

$$\sup_{\theta,\phi} \mathbb{E}_{p(\mathbf{x},y)}\big[\log \mathbb{E}_{\hat{p}(z|\mathbf{x})}\, \hat{p}(y \mid \mathbf{x}, z)\big]. \tag{3}$$

For the inner expectation, we use the Monte Carlo approximation: $\frac{1}{N}\sum_{i=1}^N \hat{p}(y \mid \mathbf{x}, z^{(i)})$, where $z^{(i)} \sim \hat{p}(z \mid \mathbf{x})$. We also use Gumbel-Softmax estimator (Jang et al., 2016; Maddison et al., 2016) to enable the learning with neural networks. The final training objective is as follows:

$$\mathcal{L}(\theta, \phi; \mathcal{D}) = \sum\nolimits_{(\mathbf{x},y)\in\mathcal{D}} \big(\log \frac{1}{N}\sum_{i=1}^N \hat{p}(y \mid \mathbf{x}, z^{(i)})\big), \tag{4}$$

where $z^{(i)}$ is sampled from $\hat{p}(z \mid \mathbf{x})$ and $\mathcal{D}$ is a dataset of samples, which could be a minibatch in practice, from the joint distribution entailed by an underlying SCM. Further details of the implementation of NCD are provided in Sec. E.3.

5. Empirical Evaluation

In this section, we evaluate our proposed method to discover contextual decomposition and CSSIs. We first consider causal systems with different types of functional model and varying complexity of

6. We assume that $Pa(Y) = \{1, \cdots, d\}$ is identifiable and correctly discovered, e.g., by some causal discovery methods.
7. In our experiments, we model $\hat{p}$ as the Gaussian and let f_θ outputs the parameters of the Gaussian distribution.

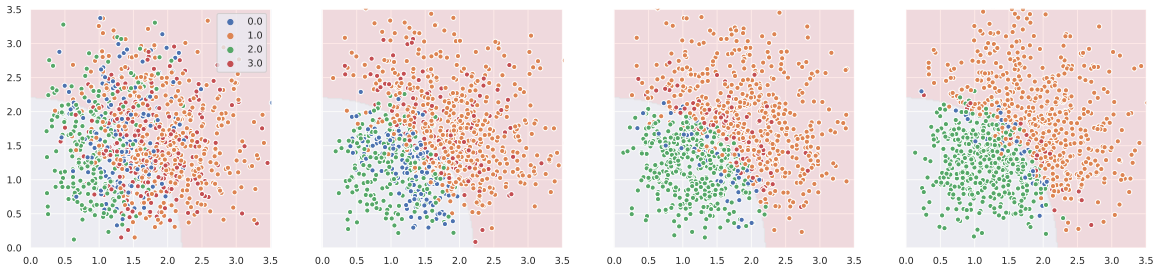


Figure 4: Learned decision boundaries (partition) for epochs 5, 10, 15, and 20. Labels 0, 1, 2, and 3 denote $z = (0, 0), (0, 1), (1, 0)$, and $(1, 1)$, respectively.

partition (Sec. 5.1), and then a more complex system reflecting the real-world physical dynamics (Sec. 5.2).

We compare our method with attention-based methods (Pitis et al., 2020) which aims to discover the local structure in a causal system of continuous variables, by computing the attention weights of the input $\mathbf{x}$. It first trains a transformer (Vaswani et al., 2017) to model the conditional distribution $p(y \mid \mathbf{x})$. Then, it uses the attention weights $(a_1, \cdots, a_d)$ of the input $\mathbf{x}$ to discover the local independence relationship. Intuitively, a low attention score a_j implies that the input variable X_j has a weak influence on the target variable Y. We also compare with a mixture model, which applies a single attention layer on top of the multiple NNs. Local independence is discovered by first computing the approximation of the Jacobian of each NNs and then applying the weighted sum with the learned attention score. For the experiments on the synthetic dataset, we let the number of the NNs be equal to the ground truth number of partition sets (i.e., oracle modeling).

For the implementation of $f_\theta(\mathbf{x} \odot z, z)$, which outputs the parameters of the Gaussian distribution $\hat{p}(y \mid \mathbf{x}, z)$, and $g_\phi(\mathbf{x})$, which outputs the parameters of the Bernoulli distributions, we used MLPs with 3 hidden layers and hidden units of 128. For all experiments, we set the batch size to 1000 and trained our method and the baselines for 100 epochs. All of our experiments were conducted with 3 different seeds, and a shaded area in a ROC plot represents a standard deviation. A detailed description of the baseline methods and further implementation details are provided in Sec. E.3.[8]

5.1. Synthetic Data

We consider causal systems where distinctive CD exists. We generate synthetic datasets consisting of the samples $(\mathbf{X}, Y)$ where $\mathbf{X} = \{X_1, \cdots, X_d\}$ is the parent variables of Y. We assume the parent variables are independent and follow the unit normal. The data generating process is as follows:

$$Y = f_i(\mathbf{X}_{A_i}, U) \quad \text{if} \quad \mathbf{x} \in \mathcal{E}_i, \quad \text{for} \quad 0 \leq i \leq N, \tag{5}$$

where U is an exogenous variable, $\{\mathcal{E}_k\}_{k=0}^N$ partitions $\mathcal{X}$, and $A_i \neq A_j$ for $i \neq j$. Here the indices enumerate the active data-generating branches; a residual cell as in Def. 9 can be introduced or relabeled separately. In our experiments, $d = 9$ and the exogenous variable follows the unit normal. We evaluate our method with varying types of functions f_k, local parents sets A_k, and partition sets $\mathcal{E}_k$. For the **functions** $\{f_k\}$, we consider nonlinear functions with (i) additive and

8. Code available at: https://github.com/iwhwang/NCD.

(ii) non-additive noise models. We use randomly initialized neural networks for the nonlinear functions, i.e., we do not assume any specific family of distributions for f_k. For the partition $\{\mathcal{E}_k\}$, we consider the cases where the **boundaries** of the partition sets are (i) linear and (ii) nonlinear. For linear boundaries, the context is selected by the largest of the branch gate scores, each computed by a linear function. The published main-text formula evaluated these scores on the local-parent blocks (e.g., $123/456/789$), whereas the formulas in Sec. E.1 use the input blocks $147/258/369$. These define different partitions in general. The original generator and run settings have not been recovered to resolve this index discrepancy; the reported curves are preserved and have not been re-evaluated. For the nonlinear boundaries, we further control the complexity of the partition by considering both cases when nonlinear boundaries are determined by the norm of $\mathbf{x}$ and some nonlinear function h. For the **local parent sets** $\{\mathbf{X}_{A_0}, \mathbf{X}_{A_1}, \mathbf{X}_{A_2}\}$, we consider two cases: (i) uniformly distributed as $\{\mathbf{X}_{123}, \mathbf{X}_{456}, \mathbf{X}_{789}\}$ and (ii) non-uniformly distributed as $\{\mathbf{X}, \mathbf{X}_{123}, \mathbf{X}_{456789}\}$. In our experiments, we use different configurations of functions, local parent sets, and boundaries. For the main experiments, we consider the case of nonlinear functions with a non-additive noise model and report the results under the additive noise model in the appendix. We provide the details of the setups and implementations in Sec. E.1.

Experimental results. We illustrate our results in Fig. 5 for (top) the uniformly and (middle) non-uniformly distributed local parent sets, respectively. Our method successfully discovers local independence structures and outperforms the baselines in most cases. We again emphasize that the mixture model exploits the ground truth number of partition sets for its implementation.

Visualization of the predicted partition. We present plots in Fig. 4 illustrating how the neural network learns the partition throughout the training procedure based on a toy experiment with 2 parent variables $\mathbf{X} = \{X_1, X_2\}$. The quality of the discovered contextual decomposition is improved as the training goes on. Our model successfully discovered the local independence and the corresponding context sets. It shows that the data points within each partition are properly classified, while the model fails to classify some points close to the decision boundary.

5.2. Complex System Reflecting Real-world Dynamics

We examine our method on the modified Spriteworld environment (Watters et al., 2019; Pitis et al., 2020), which reflects real-world physics dynamics, to evaluate our method in a more complex environment. In a 2 dimensional space, there are n moving objects (i.e., sprites), which can collide with each other. Each i-th object corresponds to two 2-dimensional variables $X_{2i-1} = (x, y)_{(i)}, X_{2i} = (v_x, v_y)_{(i)}$ which represents the position and the velocity of the object. An agent can influence the objects through an action. Action corresponds to a 2-dimensional variable $X_{2n+1} = (a_x, a_y)$, representing some position in the 2D scene, and it influences the dynamics of the nearest object, thus there are interactions between the action and each object. In each time-step, the object and action variables influence the object variables of the next time-step. Formally, $\mathbf{X} = \{X_1, \cdots, X_{2n+1}\}$ and $\mathbf{Y} = \{Y_1, \cdots, Y_{2n}\}$ where $\mathbf{X}$ denotes the object and action variables and $\mathbf{Y}$ denotes the object variables of the next time-step. Further, there exist dependencies among the parent variables. Each Y_k is generated from $\mathbf{X}$ with an unknown dynamics function f_k, i.e., $Y_k = f_k(\mathbf{X}, U)$ where U is the exogenous variable. As a whole, all $\mathbf{X}$ are involved in generating each Y_k since the objects can collide with each other and the action influences the objects. However, the interactions are sparse—the collision occurs only when objects are (i) close enough and (ii) heading toward each

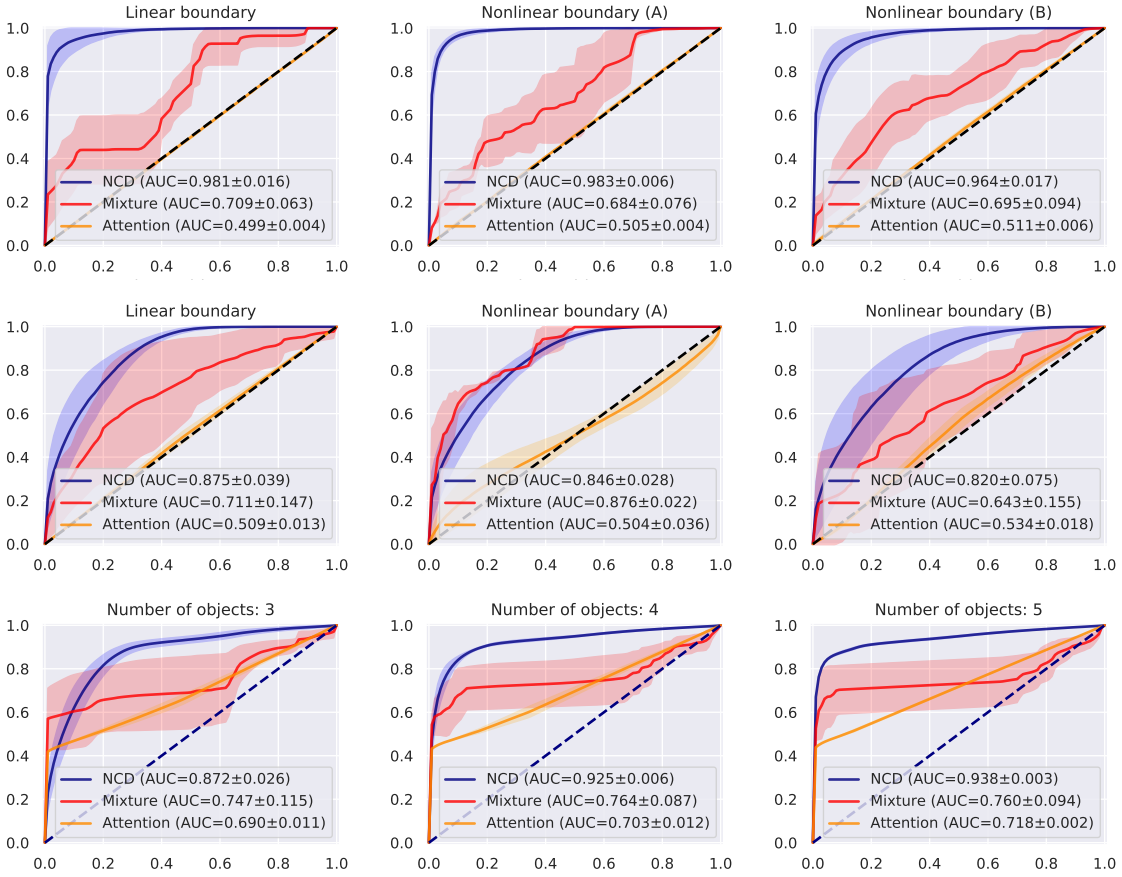


Figure 5: ROC curves for the ground truth local independence relationships on (top, middle) the synthetic dataset of (top) uniformly and (middle) non-uniformly distributed local parent sets with different types of boundaries and (bottom) on the Spriteworld.

other. Thus, the dynamics system exhibits numerous CSSI relationships. As shown in the bottom of Fig. 5, our method recovers the ground truth local structure on the varying number of objects. We provide further details of the dataset and the implementation in Sec. E.2.

6. Related Work

Local independence relationship. One of the most widely used local independence relationships is the notion of context-specific independence (CSI) (Boutilier et al., 2013; Poole, 1998). Pensar et al. (2015) proposed LDAG, a graph labeled with CSIs. In order to represent a more flexible local relationship, Pensar et al. (2016) introduced partial conditional independence (PCI), which is a generalization of CSI. The benefits of exploiting local independence relationships such as CSI has explored recently in the field of causal effect identification. Tikka et al. (2019) established the framework of causal calculus in the presence of CSI relationships along with LDAG and showed that it enables richer causal effect identification. Robins et al. (2020) proposed the algorithm for

the identification of controlled direct effect leveraging CSI relationships. While our focus is on continuous variables in SCM, Nyman et al. (2017) also considered continuous variables following Gaussian distributions but discretized the variables with certain intervals. Hence it is limited in terms of both the type of distribution and the conditioning set for local independence.

Local independence of continuous variables. Pitis et al. (2020) proposed attention-based methods to learn local causal structures. They trained a single-head transformer network and an adjacency matrix is then obtained by the product of softmax attention masks of each layer. Since it is trained *without* an inferred local causal graph, its prediction uses all of the values of the input and does not take any local independence into account for approximating conditional distribution. In contrast, our method learns *both* local causal graph within the parental relationship and conditional distribution simultaneously. Recently, Seitzer et al. (2021) proposed to estimate the conditional mutual information to discover the local independence relationships. However, they assumed that the underlying ground truth conditional distribution is the Gaussian in order to compute the conditional mutual information. In addition, their method is restricted to a single edge, i.e., one of the parent variables and the target variable, thus is not scalable and does not discover a CSSI relationships.

Neural network based causal discovery. Causal discovery attempts to reconstruct a ground truth causal graph, e.g., through conditional independence induced from the underlying system, where most of the methods fall into one of the three categories: constraint-based (Spirtes et al., 2000), score-based (Chickering, 2002; Zheng et al., 2018; Yu et al., 2019; Lachapelle et al., 2020), or hybrid (Wang et al., 2017) (see (Glymour et al., 2019) for an extensive review). In contrast, our goal is to discover local independence especially within a variable and its parents. While masking the input instances with the learned mask (i.e., adjacency matrix) is a widely used technique in prior works on neural network based causal discovery (Kalainathan et al., 2022; Ng et al., 2019; Brouillard et al., 2020), our work has a different focus. The key difference is that they learn the mask which is invariant to individual instances, where there exists a single ground truth adjacency matrix. In our work, a mask is a function of the input variables which is modeled as PIV where there exist multiple valid contextual decompositions. For the same reason, they do not require labeling instances, which is essential for valid approximation in our model.

7. Conclusion

Local independence relationships (e.g., CSI) could be leveraged in many tasks, but most of the methods are restricted to discrete variables. To this end, we formalized the notion of context-set specific independence (CSSI) providing a more flexible representation of local independence and further allowing both discrete and continuous variables. Along with CSSI, we introduced *local parent set* which allows CSSI to be represented in a causal graph and hence provides an intuitive and compact description of local independence. For the discovery of CSSIs in a system, we focused on continuous variables which is challenging, since it is impractical to directly test every individual CSSIs due to the nature of continuous variables and adopting prior methods to discover CSIs for discrete variables is not trivial. In this work, we tackled this challenge by finding the *contextual decomposition*, a partition of the joint outcome space where each partition is CSSI-inducing context-set. We presented NCD, effectively discovering such decomposition by augmenting an auxiliary partition indicator variable (PIV) which enables the gradient-based training. A PIV expresses regionwise CSSIs, while regular parent sets must additionally satisfy the minimality conditions of a contextual decomposition.

Experimental results showed that NCD successfully finds PIV and discovers CSSIs in a synthetic dataset and is also effective in a complex environment reflecting real-world dynamics.

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Appendix A. Appendix for Preliminaries

A.1. Local Independence

CSI is the most widely studied local independence relationship which generalizes the notion of conditional independence (CI). If the causal relationship between variable and its parents exhibits a CSI relationship, it implies that the variable is independent of the subset of its parents given a certain context. Labeled DAG (LDAG) (Pensar et al., 2015) is developed to encode CSI relationships on the underlying DAG, where each edge is labeled with a set of contexts, which invoke CSI relationships. Intuitively speaking, each label on its corresponding edge (j) represents the condition of the edge to be inactive, i.e., X_j does not affect Y given the context in the label. In the field of causal inference, it has been shown that the presence and knowledge of CSI relationships provide richer causal effect identification (Tikka et al., 2019). PCI (Pensar et al., 2016) extends CSI in that, given the specific context, the local independence relationship only holds in a certain domain (i.e., $\mathcal{D}_A$). We would like to note again that the line of works of aforementioned local independences mostly focused on finite discrete variables.

Revisiting Example 1, CSI relationship $Y \perp\!\!\!\perp X_2 \mid X_1 = c$ holds for every $c < 1/2$, however $Y \not\perp\!\!\!\perp X_1 \mid X_2 = c$ for any $c \in [0, 1]$. PCI relationships in this system are as follows:

$$Y \perp\!\!\!\perp X_2 \mid X_2 \in [0, 1], X_1 = c \quad (\forall c < 1/2), \tag{6}$$

$$Y \perp\!\!\!\perp X_2 \mid X_2 \in [0, 1/2c), X_1 = c \quad (\forall c \geq 1/2), \tag{7}$$

$$Y \perp\!\!\!\perp X_1 \mid X_1 \in [1/2c, 1], X_2 = c \quad (\forall c > 1/2). \tag{8}$$

Note that PCI in Eq. (6) is equivalent to CSI $Y \perp\!\!\!\perp X_2 \mid X_1 = c$. We now show that CSSI subsumes positive-probability CSI and PCI in the discrete setting of Sec. 2. CSSI itself also applies to continuous variables. A single point context of a continuous conditioning variable has probability zero and is not, by itself, a CSSI context.

Proposition 14 (CSSI subsumes CSI and PCI) *In the discrete setting, let $\mathbf{X}_A = \mathbf{a}$ have positive probability. (1) For any CSI $Y \perp\!\!\!\perp \mathbf{X}_B \mid \mathbf{X}_A = \mathbf{a}$, there is a positive-probability context $\mathcal{E}$ inducing $Y \perp\!\!\!\perp \mathbf{X}_B \mid \mathbf{X}_A, \mathcal{E}$. (2) For any PCI $Y \perp\!\!\!\perp \mathbf{X}_B \mid \mathcal{D}_B, \mathbf{X}_A = \mathbf{a}$ with $P(\mathbf{X}_A = \mathbf{a}, \mathbf{X}_B \in \mathcal{D}_B) > 0$, there is such a context $\mathcal{E}$.*

Proof (1) Take $\mathcal{E} = \{(\mathbf{a}, \mathbf{x}_B) : \mathbf{x}_B \in \mathcal{X}_B\}$. On this event, CSI gives $p(y \mid \mathbf{a}, \mathbf{x}_B) = p(y \mid \mathbf{a})$ at every supported assignment. The latter kernel depends only on the fixed A-coordinates, proving CSSI.

(2) Take $\mathcal{E} = \{(\mathbf{a}, \mathbf{x}_B) : \mathbf{x}_B \in \mathcal{D}_B\}$. The PCI equality says that $p(y \mid \mathbf{a}, \mathbf{x}_B)$ is the same for all supported $\mathbf{x}_B \in \mathcal{D}_B$. Averaging over those assignments yields $q(y \mid \mathbf{a}) = p(y \mid \mathbf{X}_A = \mathbf{a}, \mathbf{X}_B \in \mathcal{D}_B)$, which is the kernel required for CSSI. Both events have positive probability by hypothesis. ■

Appendix B. More general condition for the uniqueness of regular CSSI

We adopt and slightly modify the notion of coordinate-wise connectedness from Peters (2015).

Definition 15 *Let $\mathcal{E} \subseteq \mathcal{X}$, $\mathcal{E}|_{\mathbf{x}_A} = \{\mathbf{x}_{A^c} \mid (\mathbf{x}_A, \mathbf{x}_{A^c}) \in \mathcal{E}\} \subseteq \mathcal{X}_{A^c}$ and $proj_A(\mathcal{E}) = \{\mathbf{x}_A \mid \mathcal{E}|_{\mathbf{x}_A} \neq \emptyset\} \subseteq \mathcal{X}_A$. Let $\{\mathcal{E}|^{(k)}_{\mathbf{x}_A}\}_{k \in \mathcal{I}}$ be the path-connected components of $\mathcal{E}|_{\mathbf{x}_A}$. For $\mathbf{X}_S, \mathbf{X}_T \subseteq \mathbf{X}_{A^c}$, we*

say $\mathcal{E}|_{\mathbf{x}_A}^{(i)}$ *and* $\mathcal{E}|_{\mathbf{x}_A}^{(j)}$ *are* **directly coordinate-wise connected** *w.r.t* $\mathbf{X}_S$ *and* $\mathbf{X}_T$ *if* $proj_S(\mathcal{E}|_{\mathbf{x}_A}^{(i)}) \cap proj_S(\mathcal{E}|_{\mathbf{x}_A}^{(j)}) \neq \emptyset$ *or* $proj_T(\mathcal{E}|_{\mathbf{x}_A}^{(i)}) \cap proj_T(\mathcal{E}|_{\mathbf{x}_A}^{(j)}) \neq \emptyset$. *We say* $\mathcal{E}|_{\mathbf{x}_A}$ *is* **coordinate-wise connected** *w.r.t* $\mathbf{X}_S$ *and* $\mathbf{X}_T$ *if for any pair of its path-connected components* $\mathcal{E}|_{\mathbf{x}_A}^{(i)}$ *and* $\mathcal{E}|_{\mathbf{x}_A}^{(j)}$, *there is a sequence* $(i = a_0, a_1, \cdots, a_n = j)$ *such that* $\mathcal{E}|_{\mathbf{x}_A}^{(a_{t-1})}$ *and* $\mathcal{E}|_{\mathbf{x}_A}^{(a_t)}$ *are directly coordinate-wise connected w.r.t* $\mathbf{X}_S$ *and* $\mathbf{X}_T$ *for all* $t = 1, \cdots, n$. *We say* $\mathcal{E}$ *is* **absolutely coordinate-wise connected** *if for any* $\mathbf{X}_A \subseteq \mathbf{X}$ *and* $\mathbf{x}_A \in proj_A(\mathcal{E})$, $\mathcal{E}|_{\mathbf{x}_A}$ *is coordinate-wise connected w.r.t* $\mathbf{X}_S$ *and* $\mathbf{X}_T$ *for any arbitrary* $\mathbf{X}_S, \mathbf{X}_T \subseteq \mathbf{X}_{A^c}$.

The following proposition gives a sufficient condition for nonconvex contexts. Its openness and connectivity hypotheses apply to the same set $\mathcal{E}$; adding a null set to connect otherwise separate regions does not establish these hypotheses.

Proposition 16 (Generalization of Prop. 6) *Suppose* $\mathcal{E} \subseteq \mathcal{X}$ *is open and has positive probability, and CSSIs* $Y \perp\!\!\!\perp \mathbf{X}_{A^c} \mid \mathbf{X}_A, \mathcal{E}$ *and* $Y \perp\!\!\!\perp \mathbf{X}_{B^c} \mid \mathbf{X}_B, \mathcal{E}$ *hold. If* $\mathcal{E}|_{\mathbf{x}_{A\cap B}}$ *is coordinate-wise connected w.r.t.* $\mathbf{X}_{A\setminus B}$ *and* $\mathbf{X}_{B\setminus A}$ *for every* $\mathbf{x}_{A\cap B} \in \mathrm{proj}_{A\cap B}(\mathcal{E})$, *then* $Y \perp\!\!\!\perp \mathbf{X}_{(A\cap B)^c} \mid \mathbf{X}_{A\cap B}, \mathcal{E}$ *holds.*

Proof Write $C = A \cap B$, $R = A \setminus B$, $S = B \setminus A$, and $T = Pa(Y) \setminus (A \cup B)$. It suffices to consider a countable determining family of bounded functions ψ of Y, such as indicators of half-lines with rational endpoints. For one such ψ, let $m(\mathbf{x}) = \mathbb{E}[\psi(Y) \mid \mathbf{X} = \mathbf{x}]$. The two CSSIs give measurable functions a and b such that

$$m(\mathbf{x}) = a(\mathbf{x}_C, \mathbf{x}_R) = b(\mathbf{x}_C, \mathbf{x}_S)$$

for almost every $\mathbf{x} \in \mathcal{E}$. Strict positivity of $p(\mathbf{x})$ makes probability-null and Lebesgue-null sets equivalent inside $\mathcal{E}$.

By Fubini's theorem, for almost every fixed $\mathbf{x}_C = \mathbf{c}$ the preceding equalities hold Lebesgue-almost everywhere on the open fiber $\Omega = \mathcal{E}|_{\mathbf{c}}$. On any open product box in Ω, another application of Fubini shows that a function of $\mathbf{x}_R$ equal almost everywhere to a function of $\mathbf{x}_S$ is constant almost everywhere on that box; the coordinates T do not enter either function. Cover each path-connected component of Ω by countably many such boxes. Boxes joined by a finite chain of nonempty overlaps have the same constant, because each overlap is open and has positive Lebesgue measure. Path-connectedness therefore makes m essentially constant on each component.

If two components have overlapping R-projections, these projections are open and their nonempty intersection has positive Lebesgue measure. On each projection, $a(\mathbf{c}, \cdot)$ equals the corresponding component constant almost everywhere, so the two constants agree. The same argument applies to overlapping S-projections using b. Coordinate-wise connectedness propagates this equality to all components. This argument also covers an empty coordinate block, whose projection is a singleton in the zero-dimensional space.

Thus m is essentially constant on almost every fiber. Its constant can be taken as $\mathbb{E}[\psi(Y) \mid \mathbf{X}_C = \mathbf{c}, \mathcal{E}]$, which supplies a measurable version. Applying the argument to the countable determining family shows that the conditional distribution of Y depends only on $\mathbf{X}_C$ almost surely on $\mathcal{E}$. This is the claimed CSSI. ■

Openness is used to obtain product neighborhoods inside the context and positive-measure overlaps of component projections. A path-connected context joined only by a lower-dimensional bridge need not have these properties. Continuity of the original joint density alone does not supply neighborhoods inside an arbitrary context.

Proposition 17 (Generalization of Thm. 7) *Suppose $\mathcal{E} \subseteq \mathcal{X}$ is open, has positive probability, and is absolutely coordinate-wise connected. For any CSSI $Y \perp\!\!\!\perp \mathbf{X}_{A^c} \mid \mathbf{X}_A, \mathcal{E}$, there exists a unique $B \subseteq A$ such that $Y \perp\!\!\!\perp \mathbf{X}_{B^c} \mid \mathbf{X}_B, \mathcal{E}$ is regular.*

Proof A minimal sufficient subset exists because $Pa(Y)$ is finite. If S and T are two regular parent sets, Prop. 16 gives the CSSI with conditioning set $S \cap T$. Minimality implies $S = T = S \cap T$. In particular, the unique regular set is contained in any sufficient A. ■

A convex context of positive probability need not itself be open. Its interior is nonempty and convex, and its boundary has Lebesgue measure zero. Under the almost-sure definition of CSSI it has the same independence relationships as its interior. This yields the convex statements in the main text without requiring pointwise claims at boundary points. For nonconvex contexts, openness and coordinate-wise connectedness must be checked on one and the same representative; neither property follows merely from changing a context on a null set.

Appendix C. Omitted Proofs

Proposition 4 (CSSI Entailment) *Suppose a CSSI relationship $Y \perp\!\!\!\perp \mathbf{X}_{A^c} \mid \mathbf{X}_A, \mathcal{E}$ holds. Then, the following CSSI relationships also hold:*

(i) $Y \perp\!\!\!\perp \mathbf{X}_{B^c} \mid \mathbf{X}_B, \mathcal{E}$ for any $B \supseteq A$, (ii) $Y \perp\!\!\!\perp \mathbf{X}_{A^c} \mid \mathbf{X}_A, \mathcal{F}$ for any $\mathcal{F} \subseteq \mathcal{E}$ with $P(\mathcal{F}) > 0$.

Proof By Def. 3, there is a conditional density q depending only on $\mathbf{x}_A$ that agrees with $p(y \mid \mathbf{x})$ almost surely on $\mathcal{E}$. If $B \supseteq A$, the same q can be viewed as a function of $\mathbf{x}_B$ that ignores $B \setminus A$, proving (i). If $\mathcal{F} \subseteq \mathcal{E}$ has positive probability, any $P_{\mathbf{X}|\mathcal{E}}$-null exceptional set is also $P_{\mathbf{X}|\mathcal{F}}$-null. The same kernel therefore proves (ii). ■

Proposition 6 (Intersection Property of CSSI) *Suppose $\mathcal{E} \subseteq \mathcal{X}$ is convex and has positive probability. If CSSI relationships $Y \perp\!\!\!\perp \mathbf{X}_{A^c} \mid \mathbf{X}_A, \mathcal{E}$ and $Y \perp\!\!\!\perp \mathbf{X}_{B^c} \mid \mathbf{X}_B, \mathcal{E}$ hold, then $Y \perp\!\!\!\perp \mathbf{X}_{(A\cap B)^c} \mid \mathbf{X}_{A\cap B}, \mathcal{E}$ hold.*

Proof Positive probability and existence of a Lebesgue density imply that the convex set $\mathcal{E}$ has positive Lebesgue measure. Its interior $\mathcal{E}^\circ$ is thus nonempty and convex, and $\mathcal{E} \setminus \mathcal{E}^\circ$ is contained in its Lebesgue-null boundary. The two premise CSSIs hold on $\mathcal{E}^\circ$ by Prop. 4. Every nonempty coordinate fiber of $\mathcal{E}^\circ$ is convex and hence path-connected, so Prop. 16 applies. Its conclusion extends from $\mathcal{E}^\circ$ to $\mathcal{E}$ because the two contexts differ by a probability-null set. ■

Theorem 7 (Uniqueness of Local Parent Set) *Suppose $\mathcal{E} \subseteq \mathcal{X}$ is convex and has positive probability. There exists a unique $A \subseteq Pa(Y)$ such that the regular CSSI relationship $Y \perp\!\!\!\perp \mathbf{X}_{A^c} \mid \mathbf{X}_A, \mathcal{E}$ holds.*

Proof The finite collection of sufficient subsets of $Pa(Y)$ is nonempty because it contains $Pa(Y)$. It therefore has a minimal member. If A and B are two regular members, Prop. 6 makes $A \cap B$ sufficient. Their minimality implies $A = B = A \cap B$. ■

Theorem 10 *Let $\{(\mathcal{E}_i, A_i)\}_{i=0}^N$ and $\{(\mathcal{F}_j, B_j)\}_{j=0}^M$ be canonical CDs where $\mathcal{E}_i$ and $\mathcal{F}_j$ are open sets for all $i, j \geq 1$. Then, the following holds: (i) if $p(\mathcal{E}_i \cap \mathcal{F}_j) > 0$ then $A_i = B_j$, and (ii) for any $C \subseteq Pa(Y)$, $p(\mathcal{E}[C] \triangle \mathcal{F}[C]) = 0$[9] where $\mathcal{E}[C] = \bigcup_{A_i=C} \mathcal{E}_i$ and $\mathcal{F}[C] = \bigcup_{B_j=C} \mathcal{F}_j$.*

Proof (i) Suppose $P(\mathcal{E}_i \cap \mathcal{F}_j) > 0$ and $A_i \neq B_j$. First let $i, j \geq 1$. Without loss of generality, $A_i \setminus B_j \neq \emptyset$. The nonempty open intersection contains an open ball $\mathcal{D}$ of positive probability. Both premise CSSIs restrict to $\mathcal{D}$ by Prop. 4, and Prop. 6 gives the CSSI with parent set $A_i \cap B_j$ on $\mathcal{D}$. Since $A_i \cap B_j \subsetneq A_i$, this contradicts canonicality on $\mathcal{E}_i$.

If $i = 0$ and $j \geq 1$, the CSSI on $\mathcal{F}_j$ restricts to the positive-probability context $\mathcal{E}_0 \cap \mathcal{F}_j$. Since $B_j \subsetneq A_0 = Pa(Y)$, this contradicts canonicality on $\mathcal{E}_0$. The case $j = 0, i \geq 1$ is symmetric; when $i = j = 0$, $A_0 = B_0$ by definition. Hence $A_i = B_j$ whenever the intersection has positive probability.

(ii) For any $C \subseteq Pa(Y)$, the finite partitions give

$$P(\mathcal{E}[C] \setminus \mathcal{F}[C]) = \sum_{i:A_i=C} \sum_{j:B_j \neq C} P(\mathcal{E}_i \cap \mathcal{F}_j) = 0$$

by (i). Interchanging the partitions gives $P(\mathcal{F}[C] \setminus \mathcal{E}[C]) = 0$, proving the symmetric-difference claim. ■

Corollary 11 (Uniqueness of Distinctive Canonical CD) *Let $\{(\mathcal{E}_i, A_i)\}_{i=0}^N$ and $\{(\mathcal{F}_j, B_j)\}_{j=0}^M$ be distinctive canonical CDs where $\mathcal{E}_i$ and $\mathcal{F}_j$ are open sets for all $i, j \geq 1$. Then, the following hold: (i) $N = M$, and (ii) if $p(\mathcal{E}_i \cap \mathcal{F}_j) > 0$ then $A_i = B_j$ and $p(\mathcal{E}_i \triangle \mathcal{F}_j) = 0$.*

Proof By Thm. 10(i), any positive-probability intersection has the same parent set on both sides. In a distinctive decomposition, each parent set labels at most one cell. For every $i \geq 1$, the positive-probability set $\mathcal{E}_i$ meets some positive-probability $\mathcal{F}_j$ with $A_i = B_j$, and $j \geq 1$ because $A_i \subsetneq Pa(Y)$. Theorem 10(ii) then gives $P(\mathcal{E}_i \triangle \mathcal{F}_j) = 0$. Reversing the roles yields a bijection of the nontrivial cells, so $N = M$. The same argument applies to any positive-probability residual cell; null residuals have no positive-probability intersections. ■

Proposition 18 *For $P(\mathcal{E}) > 0$, the CSSI $Y \perp\!\!\!\perp \mathbf{X}_{A^c} \mid \mathbf{X}_A, \mathcal{E}$ holds if and only if $Y \perp\!\!\!\perp \mathbf{X}_{A^c} \mid \mathbf{X}_A, Z = 0$, where $Z = 0$ on $\mathcal{E}$ and $Z = 1$ otherwise. The latter is ordinary conditional independence under the conditional law given the discrete context $Z = 0$.*

Proof Since Z is a function of $\mathbf{X}$, the conditional distribution of Y given $\mathbf{X} = \mathbf{x}, Z = 0$ agrees with that given $\mathbf{X} = \mathbf{x}$ for $P_{\mathbf{X}|\mathcal{E}}$-almost every $\mathbf{x}$. A CSSI supplies a kernel $q(y \mid \mathbf{x}_A)$ for this law, which is exactly the stated conditional independence under $Z = 0$. Conversely, that conditional independence supplies the kernel $q(y \mid \mathbf{x}_A) = p(y \mid \mathbf{x}_A, Z = 0)$ required by Def. 3. All equalities are of conditional distributions almost surely. ■

9. $\triangle$ is a symmetric difference, i.e., $A \triangle B = (A \setminus B) \cup (B \setminus A)$

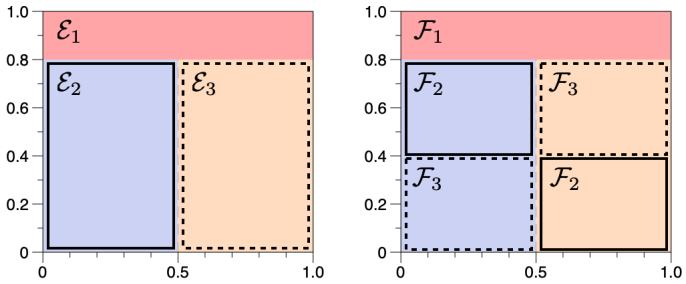


Figure 6: Example 5.

Proposition 12 (Expressing CSSI as CSI with Auxiliary Variable) *Let $\{\mathcal{E}_k\}_{k=0}^{N}$ partition $\mathcal{X}$, let $A_0 = Pa(Y)$ and $A_k \subsetneq Pa(Y)$ for $k \geq 1$, and define $Z = z_k$ when $\mathbf{X} \in \mathcal{E}_k$. On each positive-probability cell, $Y \perp\!\!\!\perp \mathbf{X}_{A_k^c} \mid \mathbf{X}_{A_k}, \mathcal{E}_k$ holds if and only if $Y \perp\!\!\!\perp \mathbf{X}_{A_k^c} \mid \mathbf{X}_{A_k}, Z = z_k$ holds. Consequently, the given pairs form a contextual decomposition if and only if these CSI relationships hold and each A_k is inclusion-minimal among their valid conditioning sets on every positive-probability cell, with $P(\mathcal{E}_k) > 0$ for $k \geq 1$.*

Proof Apply Prop. 18 separately to every positive-probability cell $\mathcal{E}_k$. Inclusion-minimality of the conditioning set is unchanged by this equivalence. The remaining requirements on A_0, the proper subsets A_k, and null residual cells are exactly those in Def. 9; they do not follow from conditional independence alone. ■

Appendix D. Additional Examples

Example 3 (Non-existence of non-trivial CD) *Let X_1, X_2 be independent $\mathrm{Unif}[0,1]$ variables, and let $U \sim \mathcal{N}(0,1)$ be independent of them. Let $Y = X_1 + X_2 + U$. In this case, there exists a unique contextual decomposition which is trivial: $\mathcal{E} = \mathcal{X}$, i.e., the causal mechanism does not exhibit any CSSI relationship.*

Example 4 (Augmented Causal Graph, Continued from Example 1) *Let X_1, X_2, and Y be defined as Example 1. Let Z be the binary random variable which is 0 if $X_1 X_2 < 1/2$ and otherwise 1. Then, the following holds:*

$$Y \perp\!\!\!\perp X_2 \mid X_1, Z = 0, \quad Y \perp\!\!\!\perp X_1 \mid X_2, Z = 1.$$

Let $\mathcal{E}_1 = \{(x_1, x_2) : x_1 x_2 < 1/2\}$, $\mathcal{E}_2 = \mathcal{E}_1^c$, and $\mathcal{E}_0 = \emptyset$. The pairs $\{(\mathcal{E}_0, \{1,2\}), (\mathcal{E}_1, \{1\}), (\mathcal{E}_2, \{2\})\}$ form a CD. The binary variable above encodes the two nonnull cells using values 0 and 1; these values are separate from the residual-cell index.

We illustrated the implementation of PIV in Figs. 1(*d*) to 1(*f*). Fig. 1(*d*) shows an augmented SCM in Example 4. Fig. 1(*e*) represents $Y \perp\!\!\!\perp X_2 \mid X_1, Z = 0$, and Fig. 1(*f*) represents $Y \perp\!\!\!\perp X_1 \mid X_2, Z = 1$.

Example 5 (Canonical CD, Illustrating Thm. 10) *Let X_1, X_2 be independent* $\mathrm{Unif}[0,1]$ *variables, and let $U \sim \mathcal{N}(0,1)$ be independent of them. Let Y be:*

$$Y = \begin{cases} X_1 + U & \text{if} \quad X_2 \geq 0.8, \\ X_2 + U & \text{if} \quad X_1 < 0.5 \text{ and } X_2 < 0.8, \\ X_2 + 2U & \text{if} \quad X_1 \geq 0.5 \text{ and } X_2 < 0.8. \end{cases}$$

Let $\mathcal{E}_1 = \{(x_1, x_2) \in (0,1)^2 \mid x_2 > 0.8\}$, $\mathcal{E}_2 = \{(x_1, x_2) \in (0,1)^2 \mid x_1 < 0.5, x_2 < 0.8\}$, $\mathcal{E}_3 = \{(x_1, x_2) \in (0,1)^2 \mid x_1 > 0.5, x_2 < 0.8\}$, and $\mathcal{E}_0 = (\mathcal{E}_1 \cup \mathcal{E}_2 \cup \mathcal{E}_3)^c$. All support and partition boundaries are assigned to this null residual. Then, the following canonical CSSIs hold:

$$Y \perp\!\!\!\perp X_2 \mid X_1, \mathcal{E}_1, \qquad Y \perp\!\!\!\perp X_1 \mid X_2, \mathcal{E}_2, \qquad Y \perp\!\!\!\perp X_1 \mid X_2, \mathcal{E}_3.$$

Therefore, $\{(\mathcal{E}_0, \mathbf{X}), (\mathcal{E}_1, \mathbf{X}_1), (\mathcal{E}_2, \mathbf{X}_2), (\mathcal{E}_3, \mathbf{X}_2)\}$ is a canonical CD. On the other hand, let $\mathcal{F}_1 = \mathcal{E}_1$, $\mathcal{F}_2 = \{(x_1, x_2) \in (0,1)^2 \mid (x_1 < 0.5, 0.4 < x_2 < 0.8)$ or $(x_1 > 0.5, x_2 < 0.4)\}$, $\mathcal{F}_3 = \{(x_1, x_2) \in (0,1)^2 \mid (x_1 < 0.5, x_2 < 0.4)$ or $(x_1 > 0.5, 0.4 < x_2 < 0.8)\}$, and $\mathcal{F}_0 = (\mathcal{F}_1 \cup \mathcal{F}_2 \cup \mathcal{F}_3)^c$. Then, the following canonical CSSIs hold:

$$Y \perp\!\!\!\perp X_2 \mid X_1, \mathcal{F}_1, \qquad Y \perp\!\!\!\perp X_1 \mid X_2, \mathcal{F}_2, \qquad Y \perp\!\!\!\perp X_1 \mid X_2, \mathcal{F}_3.$$

Therefore, $\{(\mathcal{F}_0, \mathbf{X}), (\mathcal{F}_1, \mathbf{X}_1), (\mathcal{F}_2, \mathbf{X}_2), (\mathcal{F}_3, \mathbf{X}_2)\}$ is a canonical CD. Here, $p(\mathcal{E}_2 \cap \mathcal{F}_2) > 0$ and thus $\mathbf{pa}^{\mathcal{E}_2} = \mathbf{pa}^{\mathcal{F}_2} = \mathbf{X}_2$ by Thm. 10. Also, $\mathcal{E}[\mathbf{X}_2] = \mathcal{E}_2 \cup \mathcal{E}_3 = \mathcal{F}_2 \cup \mathcal{F}_3 = \mathcal{F}[\mathbf{X}_2]$ almost surely by Thm. 10; the omitted boundaries are null.

Appendix E. Experimental Details

E.1. Synthetic datasets

The following formulas preserve the synthetic configurations described in the published appendix. Their gate input blocks differ from the published main-text formula, as explained in Sec. 5.1. The original experiment generator has not been recovered to determine which description matches the reported runs; these formulas and the numerical results have not been revalidated against that generator. The following causal system is the case of uniformly distributed local parent sets with a linear boundary. Here, g is a linear function implemented with a randomly initialized NN.

$$Y = \begin{cases} f_1(\mathbf{X}_{123}, N) & \text{if} \quad g(\mathbf{x}_{147}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})), \\ f_2(\mathbf{X}_{456}, N) & \text{if} \quad g(\mathbf{x}_{258}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})), \\ f_3(\mathbf{X}_{789}, N) & \text{if} \quad g(\mathbf{x}_{369}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})). \end{cases} \tag{9}$$

The following is the case of uniformly distributed local parent sets with a nonlinear boundary determined by the norm:

$$Y = \begin{cases} f_1(\mathbf{X}_{123}, N) & \text{if} \quad \|\mathbf{x}\| < c_1, \\ f_2(\mathbf{X}_{456}, N) & \text{if} \quad c_1 < \|\mathbf{x}\| < c_2, \\ f_3(\mathbf{X}_{789}, N) & \text{if} \quad \|\mathbf{x}\| > c_2. \end{cases} \tag{10}$$

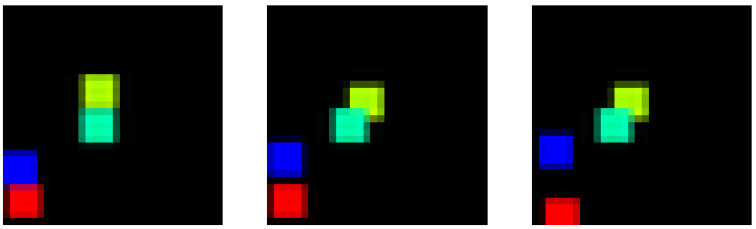


Figure 7: Example images of the Spriteworld environment.

The following is the case of uniformly distributed local parent sets with a nonlinear boundary determined by a nonlinear function g implemented with a NN:

$$Y = \begin{cases} f_1(\mathbf{X}_{123}, N) & \text{if} \quad g(\mathbf{x}_{147}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})), \\ f_2(\mathbf{X}_{456}, N) & \text{if} \quad g(\mathbf{x}_{258}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})), \\ f_3(\mathbf{X}_{789}, N) & \text{if} \quad g(\mathbf{x}_{369}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})). \end{cases} \tag{11}$$

The following causal system is the case of non-uniformly distributed local parent sets with a linear boundary. Here, g is a linear function implemented with a randomly initialized NN.

$$Y = \begin{cases} f_1(\mathbf{X}_{123}, N) & \text{if} \quad g(\mathbf{x}_{147}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})), \\ f_2(\mathbf{X}_{456789}, N) & \text{if} \quad g(\mathbf{x}_{258}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})), \\ f_3(\mathbf{X}, N) & \text{if} \quad g(\mathbf{x}_{369}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})). \end{cases} \tag{12}$$

The following is the case of non-uniformly distributed local parent sets with a nonlinear boundary determined by the norm:

$$Y = \begin{cases} f_1(\mathbf{X}_{123}, N) & \text{if} \quad \|\mathbf{x}\| < c_1, \\ f_2(\mathbf{X}_{456789}, N) & \text{if} \quad c_1 < \|\mathbf{x}\| < c_2, \\ f_3(\mathbf{X}, N) & \text{if} \quad \|\mathbf{x}\| > c_2. \end{cases} \tag{13}$$

The following is the case of non-uniformly distributed local parent sets with a nonlinear boundary determined by a nonlinear function g implemented with a NN:

$$Y = \begin{cases} f_1(\mathbf{X}_{123}, N) & \text{if} \quad g(\mathbf{x}_{147}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})), \\ f_2(\mathbf{X}_{456789}, N) & \text{if} \quad g(\mathbf{x}_{258}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})), \\ f_3(\mathbf{X}, N) & \text{if} \quad g(\mathbf{x}_{369}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})). \end{cases} \tag{14}$$

For the toy experiment for the visualization of the decision boundary, we let $X_1, X_2 \sim \mathcal{N}(0, I_3)$ and $Y = f_1(X_1)$ if $\|(X_1, X_2)\| < \epsilon$ and $f_2(X_2)$ otherwise. For the linear function g, we used a randomly initialized linear layer. For the nonlinear function g, we used a randomly initialized NN with a single hidden layer, 10 hidden units, and Tanh activation. The total number of data samples is 50000, with a ratio of 8:1:1 of training, validation, and held-out test set, respectively.

E.2. Spriteworld

While the original environment does not consider the interaction between objects, Pitis et al. (2020) modified the environment to devise the collision between moving objects and to acquire the ground truth local causal graphs. Our experiment was also conducted in the modified Spriteworld, which we denote as Spriteworld for simplicity. In this environment, there are moving objects in the scene. Following real-world dynamics, objects can collide with each other, and thus each object has an influence on others. However, their position and velocity will be affected by the others only when they collide, which occurs relatively rarely. In other words, the state of each object will only be determined by its own state in the previous time-step in most cases. Example images of the Spriteworld environment are shown in Fig. 7. In the setting of Pitis et al. (2020), each variables represent corresponding objects, i.e., $X_i = (x, y, v_x, v_y)_i$ represents the i-th object. Thus, $\mathbf{X} = \{X_1, \cdots, X_{n+1}\}$ and $\mathbf{Y} = \{Y_1, \cdots, Y_n\}$. In our experiment, we consider a more complex setup where each variable represents the position and velocity of each object, thus $\mathbf{X} = \{X_1, \cdots, X_{2n+1}\}$ and $\mathbf{Y} = \{Y_1, \cdots, Y_{2n}\}$. Further, we also include the exogenous variable.

E.3. Implementation.

Implementation details. For the implementation of our model, the distribution $p_\theta(y|x, z)$ could be any distribution as long as the log-likelihood is computable, and we chose Gaussian for simplicity. We note that ground truth $p(y|x, z)$ is non-Gaussian (i.e., non-linear function with non-additive noise model) in our experiments. For multiple Gaussian mask samples, Eq. (4) requires a mixture likelihood: the log density is $\mathrm{logsumexp}_i \log \hat{p}(y \mid \mathbf{x}, z^{(i)}) - \log N$. Averaging Gaussian random variables or their parameters is not an exact implementation of this expression. The continuous-output path of the currently linked later implementation combines samples into a single Gaussian; its correspondence to the original experimental runs has not been established. For $f_\theta(\mathbf{x}, z)$, which approximates the conditional distribution $P(Y \mid \mathbf{X}, Z)$, and $g_\phi(\mathbf{x})$, which outputs the parameters of Bernoulli distribution, we used an MLP with 3 hidden layers and 128 hidden units. For all the experiments, we set the batch size to 1000 and used the Adam optimizer with the weight decay of 10^{-5}. We set the learning rate of 10^{-2} for the synthetic dataset and 10^{-3} for the Spriteworld. Most experiments were conducted with a single RTX 3090 GPU. For the mixture model and the transformer model, we used 3 MLPs, each with 3 hidden layers and 128 hidden units, and grid-search the learning rate over $\{0.005, 0.01, 0.02, 0.03\}$.

Regularizer. One can adopt a regularizer to induce a sparse solution and prevent the model from outputting a trivial solution $z = (1, 1, \cdots, 1)$ where $z \sim g_\phi(\mathbf{x})$, e.g., adding a term $\lambda \cdot \|g_\phi(\mathbf{x})\|_1$. During our experiments, however, we empirically found that the regularizer does not significantly bring a gain. We hypothesized that since the neural network ϕ is randomly initialized, approximately half of the entries of z would be 0 at the beginning of the training, and, consequently, it would rarely converge to the trivial solution $z = (1, 1, \cdots, 1)$.

Gumbel-Softmax. Although there are other methods used for learning discrete variables, such as a REINFORCE estimator, we chose Gumbel-Softmax (Jang et al., 2016; Maddison et al., 2016) since it (i) has a lower variance compared to REINFORCE estimator, and (ii) is empirically shown to be more effective (Jang et al., 2016; Maddison et al., 2016; Ng et al., 2019). Given that the choice of reparametrization trick is not the main focus of our work, we did not compare it to other estimators.

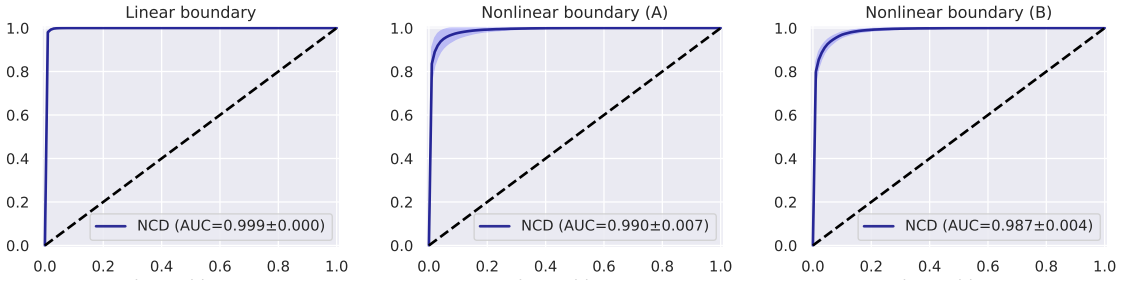


Figure 8: ROC curves for the ground truth local independence relationships on the synthetic dataset of the additive noise model.

Evaluation. We now describe the procedure for plotting the ROC curve. Let $Pa(Y) = \{1, 2, \cdots, 9\}$ (i.e., $d = 9$). Consider a *single data point* $\mathbf{x}$ where $A_{true} = \{1\}$, and the model predicted $A_{pred} = \{1, 2\}$ with a threshold τ. In this case, $TP = 1$, $FP = 1$, $FN = 0$, and $TN = 7$. We note that this evaluation procedure follows the prior work (Pitis et al., 2020). We plot the ROC curves based on *all the data points* in a test dataset. Accordingly, $TP + FP + FN + TN = B \times d$ where B is the number of test data points. Each data point belongs to one of the partitioned sets (i.e., region), and the number of data points for each region is similar. Train and test distributions are identical.

Appendix F. Additional Experiments

Here, we provide the reported results for nonlinear synthetic functions with additive noise. The gate-index ambiguity described in Sec. 5.1 and E.1 also applies to the gate-based configurations below; the plots have not been re-evaluated. We consider the following causal system with a linear boundary determined by a linear function g:

$$Y = \begin{cases} f_1(\mathbf{X}_{123}) + N & \text{if} \quad g(\mathbf{x}_{147}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})), \\ f_2(\mathbf{X}_{456}) + N & \text{if} \quad g(\mathbf{x}_{258}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})), \\ f_3(\mathbf{X}_{789}) + N & \text{if} \quad g(\mathbf{x}_{369}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})). \end{cases} \tag{15}$$

The following is the case of a nonlinear boundary determined by the norm:

$$Y = \begin{cases} f_1(\mathbf{X}_{123}) + N & \text{if} \quad \|\mathbf{x}\| < c_1, \\ f_2(\mathbf{X}_{456}) + N & \text{if} \quad c_1 < \|\mathbf{x}\| < c_2, \\ f_3(\mathbf{X}_{789}) + N & \text{if} \quad \|\mathbf{x}\| > c_2. \end{cases} \tag{16}$$

The following is the case of a nonlinear boundary determined by a nonlinear function g:

$$Y = \begin{cases} f_1(\mathbf{X}_{123}) + N & \text{if} \quad g(\mathbf{x}_{147}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})), \\ f_2(\mathbf{X}_{456}) + N & \text{if} \quad g(\mathbf{x}_{258}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})), \\ f_3(\mathbf{X}_{789}) + N & \text{if} \quad g(\mathbf{x}_{369}) = \max(g(\mathbf{x}_{147}), g(\mathbf{x}_{258}), g(\mathbf{x}_{369})). \end{cases} \tag{17}$$

As shown in Fig. 8, our method successfully discovers the local independence relationships in the additive noise system as well.

Changes

Unofficial AI-corrected version · 2026-10-25. Original: Inwoo Hwang, Yunhyeok Kwak, Yeon-Ji Song, Byoung-Tak Zhang, and Sanghack Lee, *On Discovery of Local Independence over Continuous Variables via Neural Contextual Decomposition*, CLeaR 2023, PMLR 213:448–472. Permanent original-paper link. Permanent link to this corrected version.

These proposed corrections were prepared with AI assistance. **Author review pending.** This is not an official correction issued by the publisher. Numbered locations refer to the corresponding items in the original and corrected papers.

Location	Before → After	Assumptions and limits
Def. 3; Prop. 6; Thm. 7	Pointwise CSSI → equality of conditional distributions almost surely on each positive-probability context.	Null boundaries do not determine local parents. The convex-context result is proved on the interior and extended almost surely.
App. B; Props. 16–17	Path-component connectivity on arbitrary contexts → the same connectivity hypothesis on an open context.	Openness is an additional restriction. Smooth positive densities alone do not exclude null-width bridges between positive-mass regions. Connectivity is checked on the same open representative used in the proof.
Secs. 3–4; Def. 9; Prop. 12	Unconditional trivial-regularity assertions and CD–CSI equivalence without minimality → conditional triviality statements and cellwise CSSI–CSI equivalence, with CD requirements stated separately.	Structural minimality does not imply statistical causal minimality. CSI alone does not establish regular parent sets or residual-cell regularity.
Sec. 5.1; App. E.1	Different gate-input index sets in the main and appendix descriptions → explicit distinction between those published descriptions.	The original experimental generator was not recovered, so the actual index mapping and the corresponding ROC results have not been re-evaluated.
App. A; proofs and examples	Repair the CSI/PCI subsumption proof within its discrete positive-context scope, proof wording and set-difference direction, boundary notation and local typographical errors.	These local repairs do not add a consistency or optimality guarantee for neural training.

Implementation and experiments. The currently linked implementation is shared with the later FGCDL work. Its continuous multi-mask Gaussian aggregation differs from the density mixture in Eqs. (4)–(5). The original CLeaR run configuration was not recovered, so the exposure of the reported experiments to this implementation difference is unknown. Experiments were not rerun; reported numerical values and figure assets are unchanged. Theoretical conditions are not asserted to hold for every fitted neural model or experimental system.

Citation. Cite the original paper. If this version is used, add its date and, when available, its URL to the same reference's `note`; retain the original publication year.

AI review sign-off: gpt-6.1-sol · 2026-10-25.